Chowdhury Shakil Hasan

B.Sc. in Computer Science & Engineering, Major in Information Systems
American International University-Bangladesh

+880-01576542617 chowdhuryshakilhasan@gmail.com GitHub LinkedIn

Career Objective

Final-year CSE student at American International University-Bangladesh majoring in Information Systems, with hands-on software development experience gained through an industry internship. Experienced in building responsive web applications using Next.js, React, TypeScript, Redux Toolkit, REST APIs and modern frontend technologies. Also experienced in Java Spring Boot, database-driven applications and applied machine learning. Seeking opportunities to contribute to professional software development while continuing to grow as a full-stack engineer.

Education

American International University-Bangladesh *Expected Graduation: September 2026*
Bachelor of Science in Computer Science and Engineering CGPA: 3.19 / 4.00

Experience

Sweet iTech Limited *July – August 2026*
Software Developer Intern Mohammadpur, Dhaka, Bangladesh

- Completed three end-to-end web application projects during an on-site internship with the Development Team, delivering each through GitHub and live deployment on Vercel.
- Developed responsive web applications using React, Next.js, TypeScript and Tailwind CSS, following modern frontend development practices.
- Built a role-based Product Management System using Next.js, Firebase Authentication, Cloud Firestore, Redux Toolkit and Tailwind CSS.
- Developed EasyShop, a full-featured e-commerce admin dashboard covering product, order, customer, inventory, staff, reports and analytics modules.
- Implemented state management and API data fetching with Redux Toolkit and RTK Query, and form validation with React Hook Form and Zod.
- Worked with Git-based development workflows, code review feedback, QA-reported issue fixing, reusable component design and debugging.

Research Publications

Risk-Aware Drone Routing with Stochastic Wind Fields *Dec 2025*
Co-Author — IEEE Conference, Cox's Bazar, Bangladesh ICCIT 2025 — **Published**

- Contributed to a hybrid ACO–PSO–NN framework for UAV path planning under stochastic wind conditions.
- Improved energy efficiency, travel time and CVaR-based risk through simulation-based evaluation.

Vertical Wind-Profile Driven 3D UAV Routing *Apr 2026*
Co-Author — IEEE Conference, Chattogram, Bangladesh QPAIN 2026 — **Published**

- Worked on 3D UAV routing incorporating altitude-dependent wind modelling and CVaR-based risk optimization.
- Demonstrated improvements in energy efficiency, risk reduction and flight performance under varying wind conditions.

Projects

EasyShop — E-Commerce Admin Dashboard *2026* **Live**
Next.js 16, React 19, TypeScript, Tailwind CSS, Redux Toolkit, RTK Query, React Hook Form, Zod, Recharts

- Developed a responsive e-commerce admin dashboard for managing products, orders, customers, staff, inventory and business analytics.
- Implemented product CRUD operations with search, filtering, sorting, pagination, bulk deletion and stock management.

- Built dashboard analytics with KPI cards, revenue and category charts, recent orders and low-stock monitoring using Recharts.
- Implemented Admin/Staff authentication, protected routes, role-based access, persisted sessions and Admin-only staff management.
- Added reports with date-range filtering and CSV export, along with dark mode, accessibility improvements, loading skeletons and responsive mobile navigation.
- Live · GitHub

Product Management System — Role-Based Marketplace *2026* **Live**
Next.js, Firebase Authentication, Cloud Firestore, Redux Toolkit, Tailwind CSS, Recharts

- Built a marketplace application with separate seller and customer roles, each with its own dashboard and permitted actions.
- Implemented authentication and role-based route protection using Firebase Authentication with persisted user sessions.
- Managed product listings and order data through Cloud Firestore, with global application state handled by Redux Toolkit.
- Added seller-side sales visualization using Recharts and a fully responsive interface built with Tailwind CSS.
- Live · GitHub

ParkFlow — Smart Parking Management System *2025*
Java, Spring Boot, Spring Security, JWT, MySQL, HTML/CSS, JavaScript, Bootstrap

- Developed a full-stack parking management system with role-based access (Admin/User) using Spring Boot and JWT authentication.
- Implemented secure REST APIs for slot management, vehicle tracking and booking operations with MySQL integration.
- Designed a responsive dashboard with analytics, slot filtering and real-time parking status visualization.

Technical Skills

Languages:	JavaScript, TypeScript, Python, Java, PHP, C#, C++
Frontend:	React.js, Next.js, HTML, CSS, Tailwind CSS, Bootstrap, Thymeleaf
State & Data Fetching:	Redux Toolkit, RTK Query, Fetch API, REST APIs
Forms & Validation:	React Hook Form, Zod
Backend & Frameworks:	Spring Boot, Spring MVC, Spring Security, REST API Development, JPA/Hibernate
Databases:	MySQL, Firebase Firestore
Machine Learning:	Scikit-learn, Pandas, NumPy, Unsupervised Learning, K-Means, DBSCAN, PCA
Authentication:	Firebase Authentication, JWT, OAuth2, Role-Based Access Control
Tools:	Git, GitHub, Vercel, Postman, Google Colab, Jupyter Notebook, LaTeX